

# On Developing a Hybrid Cyber-Physical Testbed for Cyber-Secure Industrial Control System

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**Abstract**—With the rapid advancement of internet technology, access to the internet has become more widespread, offering improved quality, increased availability, and reduced costs in today's era. This has paved the way for numerous devices to connect and be managed remotely via the internet, forming what is known as the Internet of Things (IoT). Similarly, industrial control systems (ICS) have also benefited from these technological advancements, enabling remote access to entire processes from anywhere in the world. Additionally, Cyber Physical Systems (CPS) have emerged, integrating smart sensing, computing, and control devices through robust and secure networks. However, this interconnectedness presents various security challenges, particularly concerning critical infrastructure (CI). We hereby investigate such security attacks in CPS, IoT and ICS and describe techniques to build resilience against them with the help of testbeds. A cost-effective prototype testbed is developed to mimic the chemical process-automation plant. The proposed testbed incorporates Virtual-Factory environment to achieve a true Hardware-In-Loop (HIL) emulation of real-world factories in cost effective manner. The entire system is controlled using the Codesys based Raspberry-pi which act as pseudo-PLC. The Modbus-TCP protocol is used to communicate between the operator Human Machine Interface (HMI) and the control system. Furthermore, we've conducted several cyber-attacks on the testbed to gather real-world data for the development of an Intrusion Detection System (IDS). These attacks include MITM-based data modification attacks utilizing the Metasploit V6 framework, and de-authentication attacks employing the aircrack-ng tool. Subsequently, the dataset obtained is utilized to train a Random Forest-based IDS model using supervised machine learning techniques.

**Keywords**— *Industrial Control System (ICS), ICS Testbed, Modbus-TCP, ICS Dataset, Intrusion Detection System (IDS), Supervised Machine Learning (ML)*

## I. INTRODUCTION

Society 5.0 builds on Industry 4.0 and is evolving into Industry 5.0, integrating humans with Cyber-Physical Cognitive Systems (CPCS). These systems combine intelligent sensing, computing [1], and control through secure networks and are widely adopted by public and private sectors managing Critical Infrastructures (CI) like power, military, water, and aviation, relying on Industrial Control Systems (ICS). Advancements in IoT have expanded CPCS applications to smart housing, agriculture, transport, e-commerce, policing, and disaster management.

With growing reliance on CIs, securing them is crucial as cyber-attacks can cause economic loss, endanger lives, and disrupt livelihoods. Notable incidents include the 2010 Stuxnet attack on Iran's nuclear facility, the 2018 WannaCrypt malware hitting Taiwan's chip plant with a \$256 million loss, and a 2021 ransomware attack costing a U.S. gas pipeline \$4.4 million [2, 4]. Strengthening resilience requires continuously identifying and addressing system vulnerabilities.

## II. BACKGROUND

The primary problem with the IoT and ICS systems is that they are highly heterogeneous systems with respect to devices, protocols, end-user applications, etc. [3, 8, 9, 10]. These IoT and ICS devices are manufactured by different Original Equipment Manufacturers (OEMs) and communicate using different protocols. The protocols used are generic and open to accommodate seamless integration to devices from different manufacturers. This makes these devices more vulnerable to cyber-attacks. The legacy systems that handle critical infrastructures still operate on very naive protocols such as Modbus-TCP, IEC61850, CAN, etc. These protocols are open and un-encrypted for ease of device integration by different OEMs [12].

Testbeds offer a practical way to strengthen IoT and ICS security by simulating attacks and analyzing network traffic to develop intrusion detection and prevention methods. They support ICS protocols like Modbus-TCP, DNP3, Profibus, and Fieldbus [2, 4], enabling realistic attack scenarios on sensors, HMI, PID setpoints, and PLC commands. By generating attack datasets, testbeds become essential for cybersecurity research. We propose studying security threats in CPCS, including IoT and ICS, and developing resilience strategies.

## III. CURRENT STATE OF ART IN TESTBEDS

We have surveyed current state-of-art testbeds which are discussed as follow.

Chen et al. [5] developed a real-time CPS testbed for smart grid communication, supporting Modbus-TCP and IEC-61850 protocols. They conducted penetration testing using MITM and DoS attacks. In their work the power system parameters were simulated using RTDS, integrated with Opnet-Modeler for network simulation and LabView PXI for instrumentation.

Wang et al. [6] developed a semi-physical testbed to simulate a boiler-turbine unit, supporting ICS protocols like Modbus-TCP, DNP3, Profibus, and Fieldbus. They conducted

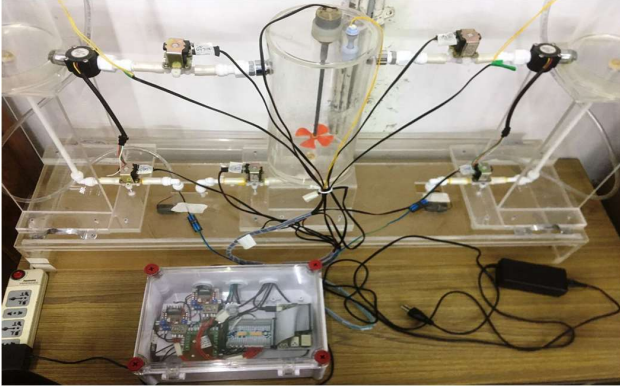


Figure 1. Chemical process automation plant prototype.

cyber-attacks targeting sensors, HMI, PID setpoints, and controller commands. The testbed, built with Raspberry Pi-based sensors and actuators, also used Raspberry Pi devices to execute attacks.

The SWaT (Secure Water Treatment) testbed [7], developed by the Singapore University of Technology and Design, mimics a six-stage water treatment plant with essential components like pumps, valves, sensors, and PLCs commonly found in ICS. It incorporates Ethernet/IP and Common Industrial Protocol (CIP) to establish a layered communication network. The dataset includes both network traffic and readings of all sensors and actuators, for benign and under attack scenario. They have created a dataset comprising of total 36 attacks, including both cyber and physical attacks assessing the vulnerabilities at physical components and communication network.

While these mentioned testbeds rely on physical or semi-physical operational systems with expensive components, low-cost architecture is a more favourable approach for academic research purposes. Our objective in this study is to construct an economical ICS prototype incorporating both physical and virtual operational systems. This approach broadens the research horizon while optimizing costs, diverging from traditional testbeds reliant on costly components.

#### IV. ATTACK SURFACE

ICS attacks often target PLCs and RTUs by intercepting function codes and altering control setpoints, causing unintended operations. Common threats include authentication bypass, MITM, and DoS attacks [5]. Additionally, attackers can

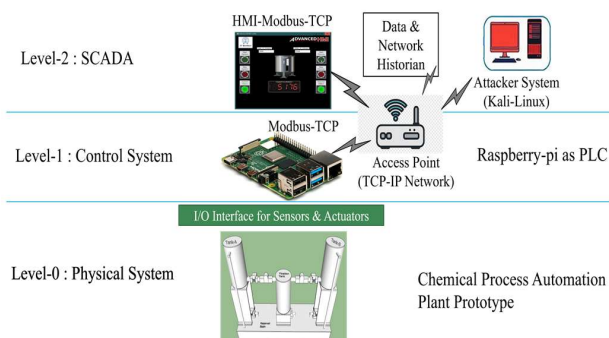


Figure 2. ICS testbed system architecture.



Figure 3. HMI user interface panel.

compromise smart sensors and actuators, logging historical data to launch replay or blocking attacks, deceiving controllers with false or fabricated data.

In ICS, Wi-Fi enables remote monitoring and control of industrial processes. Operators use it to access control systems, troubleshoot, and make adjustments. However, IEEE 802.11 de-authentication frames, often unencrypted, pose a vulnerability as they terminate connections between client devices and access points. In a de-authentication attack, the attacker exploits this vulnerability by sending forged de-authentication frames to the targeted Wi-Fi client or access point, effectively disconnecting legitimate users from the network. Subsequently, legitimate users may seek to re-establish their connection to the network, initiating the WPA2 four-way handshake. Attackers can exploit handshake traffic to intercept and crack network passwords, enabling further attacks like Evil Twin, rogue access points, MITM, and DoS.

#### V. DESIGN OF ICS TESTBED

A prototype testbed is developed to simulate the chemical process-automation plant as shown in Figure 1. The entire system is divided into three levels viz. Level-0: Physical system, Level-1: Control System, and Level-2: SCADA and HMI, as shown in Figure 2.

##### A. Level-0 Physical System

The Level-0 consists of the physical system prototype having three tanks: the left tank is pumped with liquid-A, and the right tank is pumped with liquid-B. These liquids are transferred to middle tank based on the user defined quantity in HMI and then they are mixed for process like titration for further analysis. The entire system is controlled using the Codesys based Raspberry-pi which act as pseudo-PLC. The Modbus-TCP protocol is used to communicate between the user HMI and the Raspberry-pi.

##### B. Level-1 Control System

The Codesys platform is used to program and configure Raspberry-pi to emulate the industrial PLC functionality. The Codesys IDE supports programming the PLC applications as per international industrial standard IEC-61131-3. It also provides runtime environment for monitoring and debugging the code while system is online. The Modbus-TCP modules are availed so as to make Raspberry-pi communicate to next Level-2 over Modbus-TCP protocol in TCP-IP based network.

| BSSID             | PWR | Beacons | #Data, #/s | CH | MB | ENC | CIPHER | AUTH | ESSID |
|-------------------|-----|---------|------------|----|----|-----|--------|------|-------|
| 4A:ED:0E:88:53:54 | -53 | 14      | 35         | 0  | 6  | 180 | WPA2   | CCMP | PSK   |
| 9E:53:22:24:6E:68 | -57 | 12      | 0          | 0  | 4  | 540 | WPA2   | CCMP | PSK   |
| 9C:53:22:24:6E:68 | -67 | 12      | 4          | 1  | 4  | 540 | WPA2   | CCMP | PSK   |
| 5C:EA:1D:4A:DE:87 | -79 | 10      | 0          | 0  | 6  | 65  | OPN    |      |       |
| 9C:1C:12:23:34:B1 | -80 | 11      | 0          | 0  | 11 | 130 | OPN    |      |       |
| 9C:1C:12:23:34:B0 | -80 | 12      | 0          | 0  | 11 | 130 | WPA2   | CCMP | MGT   |
| FA:8F:CA:3A:07:5B | -81 | 13      | 0          | 0  | 4  | 65  | OPN    |      |       |
| 9C:1C:12:23:34:B3 | -83 | 11      | 0          | 0  | 11 | 130 | WPA2   | CCMP | MGT   |
| 9C:1C:12:23:26:43 | -86 | 2       | 0          | 0  | 6  | 130 | WPA2   | CCMP | MGT   |

| BSSID             | STATION           | PWR | Rate  | Lost | Frames | Notes | Probes |
|-------------------|-------------------|-----|-------|------|--------|-------|--------|
| 4A:ED:0E:88:53:54 | E4:5F:01:AD:58:E7 | -1  | 0e-0  | 0    | 17     |       |        |
| 4A:ED:0E:88:53:54 | 0C:0E:76:1F:82:1A | -46 | 0e-0e | 89   | 18     |       |        |
| 9C:53:22:24:6E:68 | D8:EB:46:8A:F0:31 | -80 | 0e-0  | 204  | 26     |       |        |

Figure 4. List of access points and clients connected.

### C. Level-2 SCADA and HMI

A runtime application is developed using Advance HMI plugin for Microsoft-Visual Studio as shown in Figure 3. User can operate the system using above HMI over Modbus-TCP protocol. The HMI provided the volume of liquids to be titrated from each tank. These operational set points are communicated to the Level-1, that is, Raspberry-pi over Modbus-TCP. Further, SCADA tag historian is implemented using Ignition-SCADA by Inductive Automation. The Modbus-TCP tags are communicated to historian by pseudo-PLC at Level-1 over OPC-UA protocol.

## VI. ATTACKS SCENARIOS FOR PENETRATION TESTING

We can use this testbed to create the dataset for training the machine learning models to develop a real time Intrusion Detection System (IDS). The Modbus-TCP packet data is logged using Wireshark tool to create labelled dataset. The proposed attack scenarios are as follows:

- Data modification attack (MITM): Here the attacker can change the sensor or actuator status by sending false commands over Modbus-TCP.
- Replay attack (MITM): The attacker can replay the previous commands/network traffic to falsify the volume of liquid to be titrated/mixed.
- Syn flood attack (DoS): The attacker can flood network with redundant syn-packets making the network communication non-available
- De-authentication attack (DoS): The attacker can remove the PLC or HMI from the Wi-Fi network with the help of de-authentication frames.
- Program modification (Authentication): The attacker can get unauthorized access to the PLC and will upload wrong/malicious program to the PLC.

### A. Attack on Modbus-TCP

Out of above proposed attacks we have implemented the Data modification attack using Metasploit V6 framework. The labelled dataset is created using the network packet capture, which are labelled based on known IP of attacker and benign devices. Using Metasploit V6 data-modification and data-sniffing attack were implemented while system was operating in normal mode. This dataset was used to train a Random-Forest based IDS model using supervised machine learning technique with accuracy of 100%. Although these results are obtained with a very skewed distribution of benign and attack packets. To remedy this, we will be scaling up the system with multiple



Figure 5. HMI interface after de-authentication attack.

pseudo-PLCs which will be integrated with virtual factory-IO simulator, as described in Section VII.

### B. De-authentication Attack and Dataset

We mounted de-authentication attack on the ICS testbed which disconnected the Raspberry-pi based pseudo-PLC from the network. Here aircrack-ng tool suite is used to capture the packets in the network and monitor them. By monitoring the network traffic in its vicinity, attacker can get the MAC address of access point and client device (Raspberry-pi based pseudo-PLC here). Corresponding observation by attacker is shown in Figure 4, this displays the list of detected access points and also a list of connected clients. Further, attacker can inject packets into the network using aircrack-ng. Leveraging this attacker forcefully sends de-authentication frames to access point by spoofing the MAC address of client device (Raspberry-pi based pseudo-PLC here) leading to de-authentication attack. It sends total of 128 packets between access point and client, out of which 64 packets are sent to access point and other 64 packets are sent to client. Before sending the de-authentication packets, the operator monitoring the control system, was able to observe the state of the process as shown in the Figure 3. However after de-authentication attack, the connection was terminated and operator can no longer monitor the system resulting into DoS attack and corresponding error message is displayed on HMI as shown in Figure 5. This will force the Raspberry-pi to re-establish the connection with access point by initiating the WPA2 handshake. Further, Figure 6 shows the network traffic captured by Wireshark tool. De-authentication frame control field 0xc000 and the corresponding source and destination address of the packet is captured. Then, attacker leverages this opportunity and captures this handshake and crack the network password using dictionary attack as shown in Figure 7. By cracking the password, attacker can get into the network and

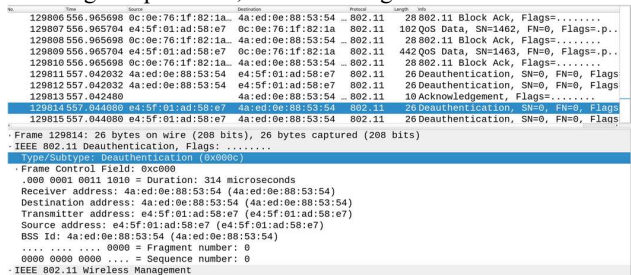


Figure 6. De-authentication packets captured by Wireshark.



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Aircrack-ng 1.6
[00:00:00] 4270/10303727 keys tested (9095.68 k/s)
Time left: 18 minutes, 52 seconds          0.04%

KEY FOUND! [ alohomora ]

Master Key   : B8 BE 27 2B A1 E9 91 F1 19 0C 63 EB E0 23 AC A0
              FA 6E 89 7F CD 56 75 03 96 CB A3 A6 DF 2E 06 F2

Transient Key : 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
              00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
              00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
              00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

EAPOL HMAC   : 7D AF 6E F1 32 D0 C8 3F 59 3F 00 3B 32 D8 1B B7

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Figure 7. Network password cracked using aircrack-ng.

perform other attacks such as MITM, Evil Twin attack, rogue access point attack. We are in the process of mounting these attacks and create corresponding dataset to design a reliable IDS.

## VII. TESTBED WITH VIRTUAL FACTORY ENVIRONMENT

A 3D visualization of virtual factory with industry grade sensors, actuators and controllers is possible with software Factory I/O developed by M/s. RealGames [11]. It provides integration of these virtual sensors and actuators with real controllers such as Raspberry-pi. A pseudo-PLC implemented using Codesys software can be integrated with this virtual factory environment. The Figure 8 shows the overall virtual environment setup using Factory I/O, Codesys and Wireshark to capture the network traffic. Here, Codesys is used to load control program using ladder logic language on Raspberry-pi, which acts as pseudo-PLC. The various sensor and actuator tags are mapped using Modbus-TCP protocol. Thus, the sensor data is read in real time over Modbus-TCP and corresponding control action is to be taken is written to actuators, also using Modbus-TCP. This gives true Hardware-in-the-loop (HIL) simulation of industrial automation system. This setup can be used for penetration testing for various industrial communication protocols viz MODBUS-TCP, Siemens Profibus and ProfiNet.

## VIII. SUMMARY AND FUTURE CHALLENGES

An economical testbed incorporating both physical and virtual OT systems is proposed for realistic dataset creation for analysing and detecting attacks on IoT and ICS network. We are currently working on integrating both physical and virtual systems as shown in Figure 9, so as to create a balanced and heterogenous dataset. This deployment can be used for creating further scenarios for penetration testing with MITM and replay

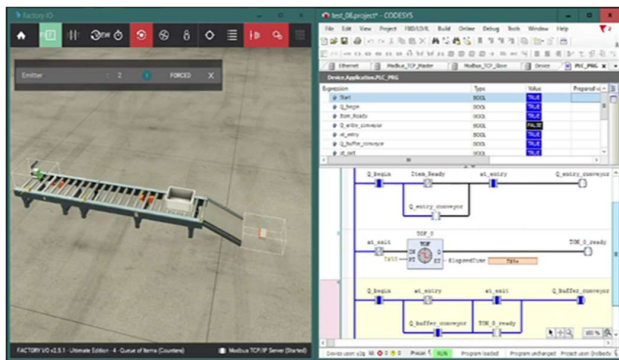


Figure 8. Factory I/O simulator with Codesys pseudo-PLC.

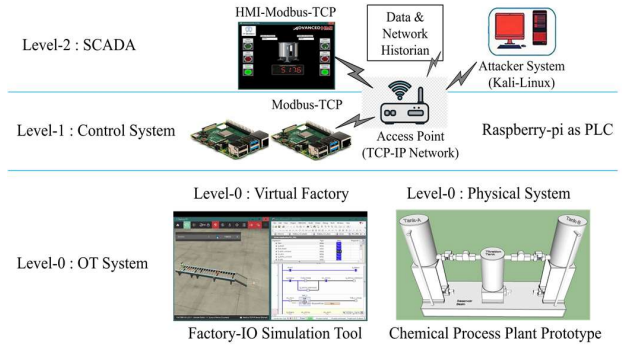


Figure 9. Proposed testbed with virtual factory and ICS prototype.

attacks on Modbus-TCP based communication. The architecture will be expanded to incorporate more IoT and ICS devices to get more heterogenous device and communication profile of the network traffic. The Factory I/O simulator along with Codesys based pseudo-PLC will be used to create various attack scenarios for ICS and corresponding dataset will be captured for further analysis. Further, we propose to validate the Raspberry-pi based control system with commercial PLC and SCADA based control.

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